Exploring Effects of Interactive VR Sensory Environment on Anxiety Reduction in Adolescents with Autism: A Pilot Study

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Introduction

- **Autism Spectrum Disorder (ASD)** is a neurological and developmental disability often characterized by [2]:
  - Social and communicative impairments
  - Restricted and repetitive behaviors
- Sensory processing difficulties coexist with anxiety in individuals with ASD [1].
- Children/adolescents with ASD are more likely to experience higher levels of anxiety than neurotypical individuals [3].
- **Multi-sensory environments (MSEs)** provide people with developmental disabilites with a range of sensory equipment, toys and activities.
- VR-based sensory rooms are more affordable, accessible (portable), and individualized.
- Need for further developed anxiety management techniques and coping strategies for patients with ASD.

Objective

- Investigate how various auditory, visual, and interactive components contribute to user satisfaction and sensory-related anxiety reduction in a VR sensory environment.

Methodology

Technical Development

- Unity3D Game Engine + Meta Interaction SDK
- Meta Quest 2 Headset
- ***Activities*** - Bubble Popping, Color Changing Light Panel, 3D Object Manipulation, Alphabet Board, Glowing Stars
- ***Visual/Auditory Elements*** - Text instructions, video demos, action-based sound effects, ambient audio music (optional), dim mood lighting

Study Protocol

- 7 participants between the ages of 21 and 44 (3 females, 4 males)
- 11-question anxiety questionnaire adapted from the Glasgow Anxiety Scale for People with Intellectual Disabilities
- 5-point numeric rating scale evaluated satisfaction and anxiety reduction of each element

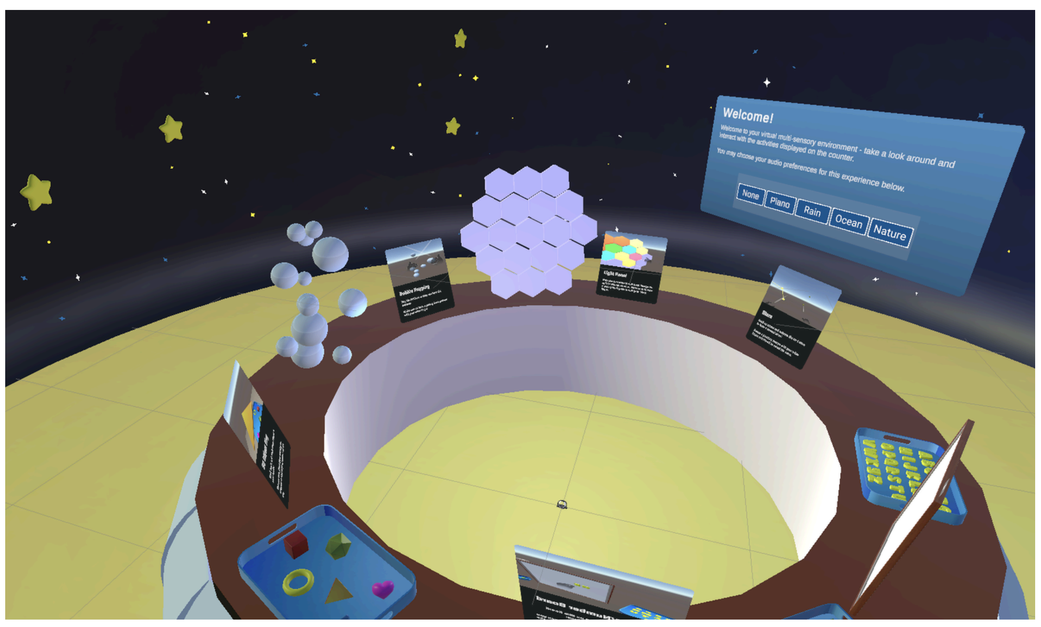


Figure 1. Full view of virtual environment

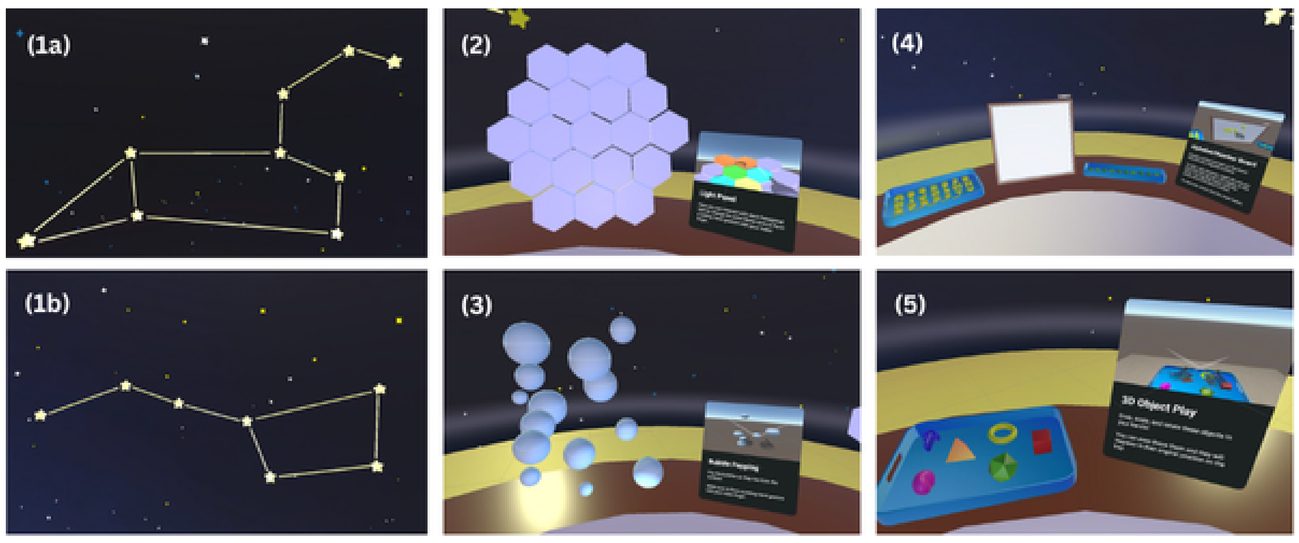


Figure 2. Five interactive activities present within the virtual environment: (1a) Leo Constellation, (1b) Big Dipper Constellation, (2) Light Panel, (3) Bubble Popping, (4) Alphabet and Number Board, (5) 3D Object Play.

Results & Discussion

Table 1. Mean, standard deviation, and difference of pre- and post- intervention scores, *p*-value, effect size (η^2), and average duration of use.

	Pre		Post					
	Mean	**SD**	**Mean**	**SD**	**Difference**	**Sig.**	**Partial Eta Squared**	**Duration of Use**
Measure (n=7)	7.43	3.83	2.00	1.83	5.43	0.008	0.721	5 min 15 sec

	Mean Rating
3D Objects	4.57
Sound Effects	4.57
Bubbles	3.71
Text Instructions	3.71
Constellations	3.57
Ambient Audio	3.57
Light Panel	3.43
Video Demos	3.43

Table 2. Average ratings of features' contribution towards satisfaction with environment

	Mean Rating
3D Objects	4.43
Bubbles	4.14
Light Panel	3.86
Constellations	3.57
Sound Effects	3.71
Ambient Audio	3.57
Text Instructions	3.14
Video Demos	3.14

Table 3. Average ratings of features' contribution towards anxiety reduction

- **Average Overall Satisfaction Rating**: 4.57
- **Average Anxiety Reduction Effectiveness Rating**: 4.23
- Sound effects, 3D shapes, and bubbles had the greatest contribution towards user satisfaction and anxiety reduction (Table 2 & 3)
- Reported an appreciation for originality, creativity and abundance of choices.
- Had many recommendations for improving the lighting, audio and visual displays to increase comfortability.

References

[1] Shelly J. Lane, Stacey Reynolds, and Levent Dumenci. 2012. Sensory over-responsivity and anxiety in typically developing children and children with autism and attention deficit hyperactivity disorder: Cause or coexistence? The American Journal of Occupational Therapy 66, 5 (Sep 2012), 595–603. https://doi.org/10.5014/ajot.2012.004523

[2] Catherine Lord, Edwin H Cook, Bennett L Leventhal, and David G Amaral. 2000. Autism spectrum disorders. Neuron 28, 2 (Nov 2000), 355–363. https://doi.org/10.1016/s0896-6273(00)00115-x

[3] Bonnie M. MacNeil, Vicki A. Lopes, and Patricia M. Minnes. 2009. Anxiety in children and adolescents with autism spectrum disorders. Research in Autism Spectrum Disorders 3, 1 (Jan 2009), 1–21. https://doi.org/10.1016/j.rasd.2008.06.001